

Hoofed mammals

PHYLUM	Chordata
CLASS	Mammalia
ORDER	Perissodactyla, Artiodactyla
FAMILIES	13
SPECIES	399

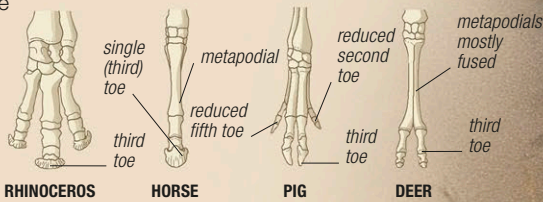
CLASSIFICATION NOTE

Hoofed mammals are classified in 2 orders: Perissodactyla (odd-toed hoofed mammals), and Artiodactyla (even-toed hoofed mammals). Although superficially similar, the 2 orders are not closely related. However, they are often grouped together because they share a range of common traits. There is evidence to suggest that the closest relatives of the artiodactyls are the cetaceans (whales and relatives), and some taxonomists now group them together in the order Cetartiodactyla.

Hoofed mammals are a highly successful group. Their position as the dominant terrestrial herbivores can be attributed largely to their speed and endurance (they are able to outrun most predators), and to the fact that they are well equipped to break down the cellulose in their plant diet. Despite a variety of body forms, most species have a long muzzle, a complex battery of grinding teeth, and a barrel-shaped body. The group consists of odd-toed hoofed mammals (such as tapirs) and even-toed hoofed mammals (such as deer). Wild odd-toed hoofed mammals are found in Africa, Asia, and South and Central America. Wild even-toed hoofed mammals are distributed worldwide (except the West Indies, Australasia, and Antarctica). Hoofed mammals are mostly found in open habitats, such as savanna. Domestic hoofed mammals are found almost anywhere there are humans.

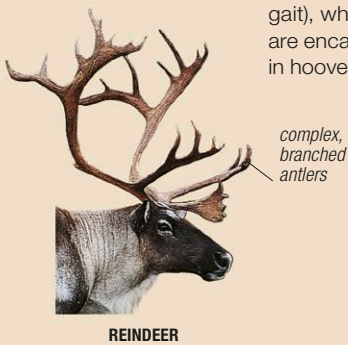
Anatomy

The ability of hoofed mammals to run swiftly for long distances is largely due to the structure of the limbs, which are adapted for simple but powerful forwards and backward movement. Each limb is embedded in the body wall as far down as the elbow or knee joint. Below this joint are the radius and ulna (front limbs) or tibia and fibula (back limbs), then the greatly elongated metapodials (the palm and foot bones in humans). This longer lower limb (and increased movement in the shoulder joint) gives a longer stride length and hence more speed. These animals also have a reduced number of toes, which means fewer muscles and tendons and therefore lower energy demands (which aids endurance). They run on their toes (unguligrade gait), which are encased in hooves.



ODD AND EVEN TOES

In odd-toed hoofed mammals (rhinoceros and horse), the leg's weight rests on the central (third) toe. In even-toed hoofed mammals (pig and deer) the weight is borne by the third, and fourth toes; the second and fifth toes are greatly reduced (pig) or lost (deer).



REINDEER



GOAT

HORNS AND ANTLERS

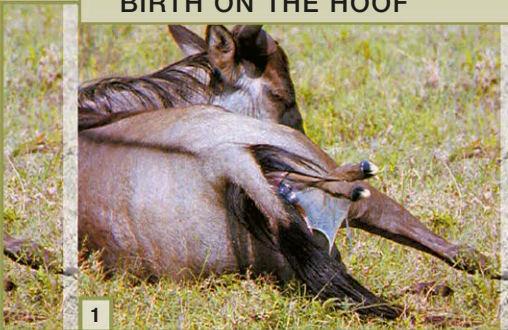
Antlers are bony outgrowths of the skull that are shed every year. Horns are permanent and have a bony core covered in keratin. Giraffe horns (ossicones) are covered with skin and develop from cartilage that later ossifies and fuses with the skull.

ESCAPING DANGER

Hoofed mammals must be able to detect danger quickly to survive. To do this they have mobile, tubular ears and acute hearing, an excellent sense of smell, and eyes on the side of the head, giving all around vision. When frightened, they flee at great speed. Antelopes, such as these impala, often make spectacular leaps as they escape.



BIRTH ON THE HOOF



1

THE BIRTH BEGINS

During labor, this female wildebeest is vulnerable and restless, alternately lying down and standing up.



2

THE HEAD APPEARS

The calf is born head first, but the long front legs are first into the outside world.



3

A NEW LIFE

The newborn wildebeest emerges, partially covered by the amniotic membrane.



4

SYNCHRONIZED CALVING

Most wildebeests calve in the same 3-week period so that, despite some losses, many calves survive.